Hypothesis Testing

Starter

1) $X \sim B(20, 0.4)$. Calculate:

a) $P(X = 5)$

b) $P(X = 10)$

c) $P(X \leq 2)$

d) $P(X \geq 18)$

1a

1b

1c

1d

2) Wanda rolls a fair die eight times.

a) Suggest a suitable model for the random variable X, the number of times the die lands on five.

2a

b) Calculate:

i. $P(X = 2)$

ii. $P(X \geq 4)$

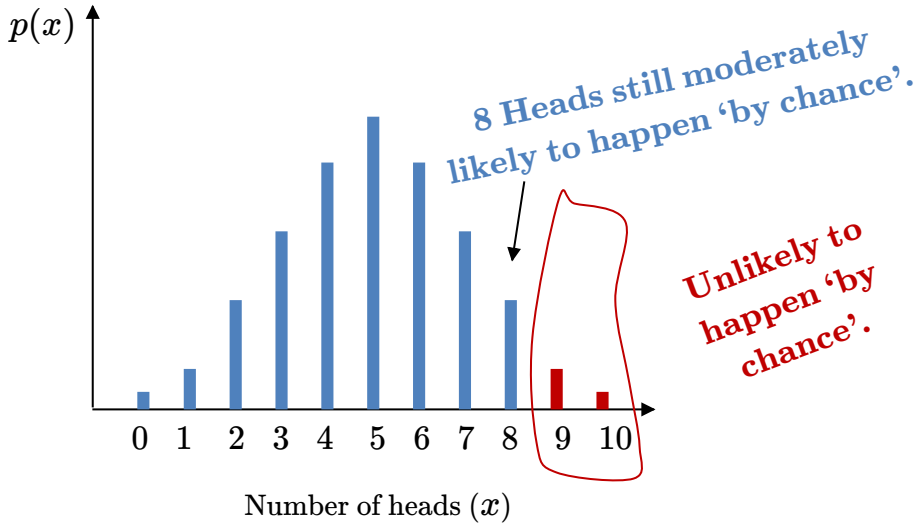
2b(i)

2b(ii)



I throw a coin 10 times. For what numbers of heads might you conclude that the coin is biased towards heads? Why?

Our intuition is that the further away we are from the 'expected' number of heads (i.e. 5 heads out of 10), the more unlikely it is.





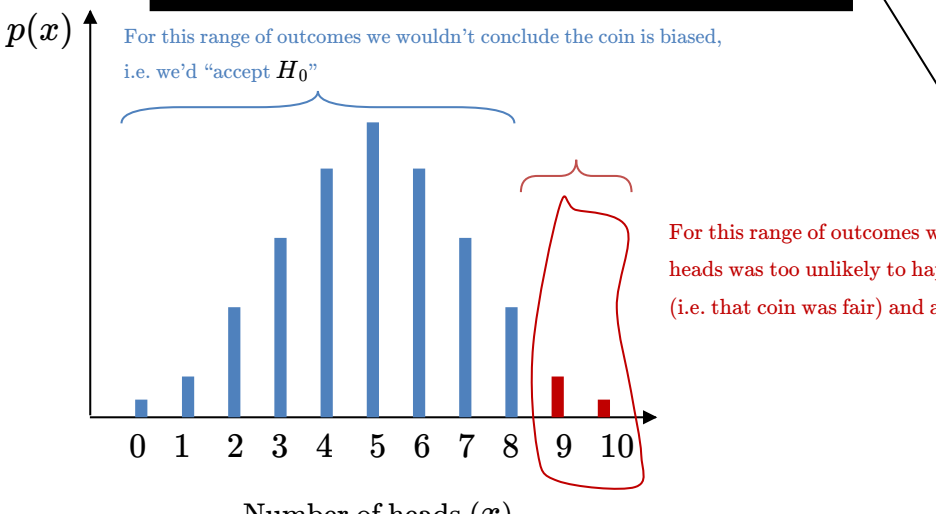
I throw a coin 10 times. For what numbers of heads might you conclude that the coin is biased towards heads? Why?

A hypothesis is a statement made about the value of a **population parameter** that we wish to test by collecting evidence in the form of a sample.

! The **null hypothesis**, H_0 is the default position, i.e. that nothing has changed, unless proven otherwise.

! The **alternative hypothesis**, H_1, is that there has been some change in the population parameter.

In this context...



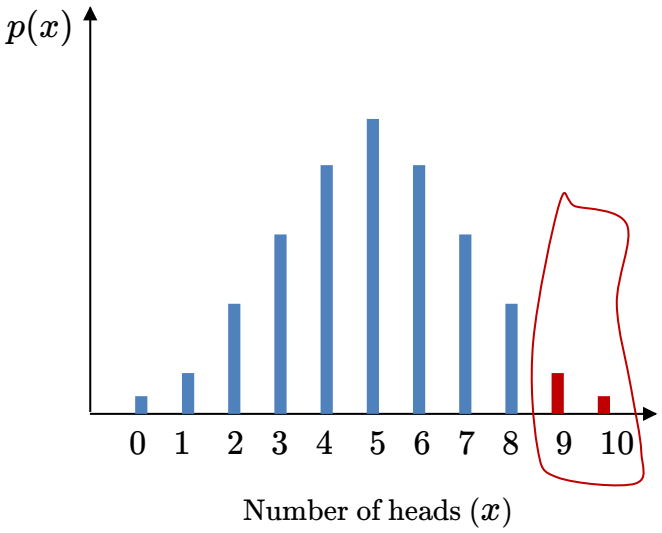
For this range of outcomes we'd conclude that this number of heads was too unlikely to happen by chance, and hence reject H_0 (i.e. that coin was fair) and accept H_1 (i.e. that coin was biased).



I throw a coin 10 times. For what numbers of heads might you conclude that the coin is biased towards heads? Why?

In this context...

In a hypothesis test, the evidence from the sample is a **test statistic**.

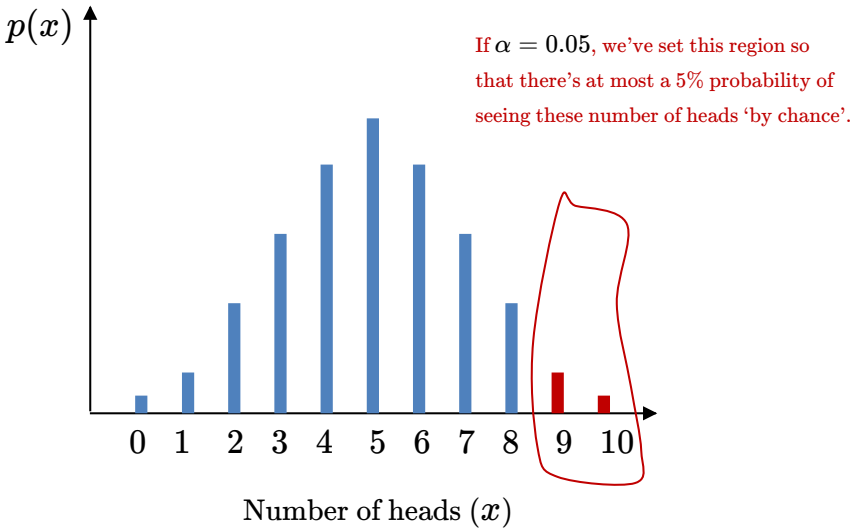




I throw a coin 10 times. For what numbers of heads might you conclude that the coin is biased towards heads? Why?

In this context...

! The **level of significance** α is the maximum probability where we would reject the null hypothesis.
This is usually 5% or 1%.



Hypothesis testing in a nutshell then is:

1. We have some hypothesis we wish to see if true (e.g. coin is biased towards heads), so. . .
2. We collect some sample data by throwing the coin (giving us our 'test statistic') and. . .
3. If that number of heads (or more) is sufficiently unlikely to have emerged 'just by chance', then we conclude that our (alternative) hypothesis is correct, i.e. the coin is biased.



[Textbook] John wants to see whether a coin is unbiased or whether **it is biased towards coming down heads**. He tosses the coin 8 times and counts the number of times X, it lands head uppermost.

We said that our two hypotheses are about the population parameter.

Suppose p is the probability of a coin landing heads.

Null hypothesis:

Alternative hypothesis

Under the **null hypothesis** H_0, we **assume that the population parameter is correct**, in this case, that it is a normal coin and the probability of Heads is 0.5

Under the **alternative hypothesis** H_1, there has been an underlying change in the population parameter, in this case that the coin is actually biased towards Heads.

The latter is known as a '**one-tailed test**' because we're saying the coin is biased one way or the other (i.e. $p > 0.5$ or $p < 0.5$).
But we could also have had the hypothesis 'the coin is biased (either way)', i.e. $p \neq 0.5$. This is known as a **two-tailed test**.

Further Example 9

[Textbook] An election candidate believes she has the support of 40% of the residents in a particular town. A researcher wants to test, at the 5% significance level, whether the candidate is over-estimating her support. The researcher asks 20 people whether they support the candidate or not. 3 people say they do.

a) Write down a suitable test statistic.

b) Write down two suitable hypotheses.

c) Explain the condition under which the null hypothesis would be rejected.

a

For a hypothesis test involving the binomial distribution, the test statistic is always the **count of successes**.

b

The alternative hypothesis is that the candidate is **overestimating** her support, so we're interested where **less than 40%** support them (more than 40% would not undermine the candidate's claim).

c

This is the hard bit!

We always calculate the probability of seeing this outcome **or more extreme** (in this case, 'more extreme' meaning even fewer the 3 people, because this takes us even further from the expected number of people out of the 20 (i.e. 8) who would support them.

The "$p = 0.4$" bit is because, as discussed before, we calculate the probability of seeing the observed outcome of 3 people (or more extreme) if it occurred **purely by chance** (the null hypothesis), i.e. if the candidate **did** have 40% support.

Test Your Understanding 10

In the UK, 5% of students turn up late to school each day. Miss Weir wishes to determine, to a 10% significance level if her school has a problem with attendance. She stands at the front gate one day and finds that 6 of the 40 students who pass her are late.

a) Write down a suitable test statistic.

b) Write down two suitable hypotheses.

c) Explain the condition under which the null hypothesis would be rejected.

a

b

If there is indeed a problem with attendance (i.e. the school is not the norm) then we expect the proportion p of students late at Tiffin to be higher than 5%.

c

Just to reiterate again what's going on here:

a) We assume that the school is the norm (i.e. 5% of students are late), and calculate the probability that 6 or more students would be late under this assumption.

b) If the probability of this happening by chance is sufficiently unlikely (less than the 10% significance level), we conclude that it probably isn't just by chance, and the school's lateness rate is worse than 5%.

1. (a) Explain what a hypothesis test is. (b) Define a null and an alternative hypothesis and state their symbols. (c) Define a test statistic.
2. Classify each test as one- or two-tailed:
(a) $H_0 : p = 0.8, H_1 : p > 0.8$;
(b) $H_0 : p = 0.6, H_1 : p \neq 0.6$;
(c) $H_0 : p = 0.2, H_1 : p < 0.2$.

1a

1b

1c

2a

2b

2c

3. Dmitri tests whether a die is biased towards six. He rolls it 60 times, counting sixes.
(a) Describe the test statistic.
(b) Write a suitable null hypothesis.
(c) Write a suitable alternative hypothesis.

a

b

c

4. Shell tests whether a coin is biased. She tosses it 100 times and counts heads. She says the test statistic is the probability of heads.
(a) Explain her mistake and give the correct statistic.
(b) Give a null hypothesis.
(c) Give an alternative hypothesis.

a

b

c

5. Historically 10% of articles are faulty. In a sample of 100 from a new process, 8 are faulty. Test at the 5% level whether the proportion has fallen.
(a) Suggest a test statistic.
(b) State suitable hypotheses.
(c) Explain when the null hypothesis would be rejected.

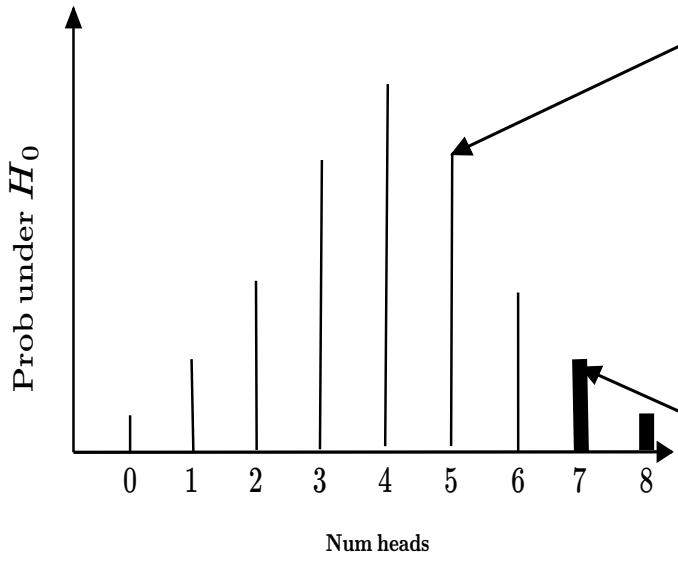
a

b

c

John wants to see whether a coin is unbiased or whether **it is biased towards coming down heads**. He tosses the coin 8 times and counts the number of times X, it lands head uppermost. **What values would lead to John's hypothesis being rejected?**

As before, we're interested how likely a given outcome is likely to happen 'just by chance' under the null hypothesis (i.e. when the coin is not biased).



The probability of exactly five heads is only 22%. Why would seeing five heads be insufficient evidence of bias?

However, there are values which collectively form a range of 'extreme values' where it would be unlikely that the coin would be unbiased. Their combined probability is limited by the level of significance set (e.g. 5%)

John wants to see whether a coin is unbiased or whether **it is biased towards coming down heads**. He tosses the coin 8 times and counts the number of times X, it lands head uppermost. **What values would lead to John's hypothesis being rejected**, if the significance level was 5%?

What's the probability that we would see **6 heads**, or an **even more extreme value**? Is this sufficiently unlikely to support John's claim that the coin is biased?

What's the probability that we would see **7 heads**, or an **even more extreme value**?

C.D.F. Binomial table: $p = 0.5, n = 8$

x	$P(X \leq x)$
0	0.0039
1	0.0352
2	0.1445
3	0.3633
4	0.6367
5	0.8555
6	0.9648
7	0.9961

Critical Regions and Values 17

John wants to see whether a coin is unbiased or whether **it is biased towards coming down heads**. He tosses the coin 8 times and counts the number of times X, it lands head uppermost. **What values would lead to John's hypothesis being rejected**, if the significance level was 5%?

✎ The **critical region** is the range of values of the test statistic that would lead to you rejecting H_0

If level of significance 5%, critical region?

✎ The value(s) on the boundary of the critical region are called **critical value(s)**.

Critical value:

C.D.F. Binomial table: $p = 0.5, n = 8$

x	$P(X \leq x)$
0	0.0039
1	0.0352
2	0.1445
3	0.3633
4	0.6367
5	0.8555
6	0.9648
7	0.9961
8	1

Tip: Use the first value AFTER the one in the table that exceeds 95%.

Quickfire Critical Regions 18

Determine the critical region when we throw a coin where we're trying to establish if there's the specified bias, given the specified number of throws, when the level of significance is 5%.

Coin thrown 5 times. Trying to establish if biased towards heads.

$p = 0.5, n = 5$

x	$P(X \leq x)$
0	0.0312
1	0.1875
2	0.5000
3	0.8125
4	0.9688

Critical region:

Coin thrown 10 times. Trying to establish if biased towards heads.

$p = 0.5, n = 10$

x	$P(X \leq x)$
0	0.0010
1	0.0107
2	0.0547
...	...
7	0.9453
8	0.9893
9	0.9990

Critical region:

Coin thrown 10 times. Trying to establish if biased towards tails.

$p = 0.5, n = 10$

x	$P(X \leq x)$
0	0.0010
1	0.0107
2	0.0547
...	...
7	0.9453
8	0.9893
9	0.9990

Critical region:

Reminder: At the positive tail, use the value AFTER the first that exceeds 95% (100 - 5).

At the negative tail, we just use the first value that goes under the significance level.

Actual Significance Level 19

John wants to see whether a coin is unbiased or whether **it is biased towards coming down heads**. He tosses the coin 8 times and counts the number of times X, it lands head uppermost. **What values would lead to John's hypothesis being rejected**, if the significance level was 5%?

We saw earlier that the critical region was $X \geq 7$, i.e. the region in which John would reject the null hypothesis (and conclude the coin was biased).

We ensured that $P(X \geq 7)$ was less than the significance level of 5%.
But what actually is $P(X \geq 7)$?

This is known as the actual significance level, i.e. the probability that we're in the critical region. We expected this to be less than, but close to, 5%.

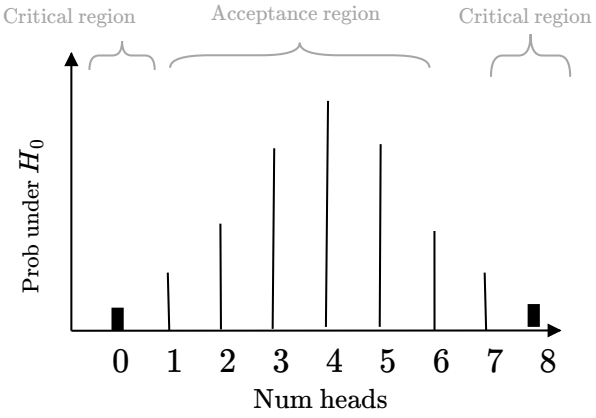
✎ The **actual significance level** is the actual probability of being in the critical region.

C.D.F. Binomial table: $p = 0.5, n = 8$

x	$P(X \leq x)$
0	0.0039
1	0.0352
2	0.1445
3	0.3633
4	0.6367
5	0.8555
6	0.9648
7	0.9961
8	1

Two-tailed test 20

Suppose I threw a coin 8 times and was now interested in how may heads would suggest it was a **biased coin** (i.e. either way!). How do we work out the critical values now, with 5% significance?



Critical region at positive tail:

Critical region at negative tail:

C.D.F. Binomial table: $p = 0.5, n = 8$

x	$P(X \leq x)$
0	0.0039
1	0.0352
2	0.1445
...	...
6	0.9648
7	0.9961
8	1

Test Your Understanding 21

A random variable X has binomial distribution $B(40, p)$. A single observation is used to test $H_0 : p = 0.25$ against $H_1 : p \neq 0.25$.

a) Using the 2% level of significance, find the critical region of this test. The probability in each tail should be as close as possible to 0.01.

b) Write down the actual significance level of the test.

The $\neq$ indicates bias either way, i.e. two-tailed.

This means you find the closest to 0.01 (even if slightly above) rather than the closest under 0.01

a

b

To ensure all method marks always show the probability of being in the critical region (even if you don't subsequently need the value!)

Note that X can't go below 0 or exceed 40.

C.D.F. Binomial table: $p = 0.25, n = 40$

x	$P(X \leq x)$
2	0.0010
3	0.0047
4	0.0160
5	0.0433
16	0.9884
17	0.9953
18	0.9983
19	0.9994

Warning: Textbook has several typos in this example.

Exercise 7B - Statistics 1 Pages 103-105 22

2. For $X \sim B(10, p)$, find the 5% critical region for $H_0 : p = 0.2$ against $H_1 : p > 0.2$.
3. A single observation from $B(20, p)$ is used to test $H_0 : p = 0.15$ against $H_1 : p < 0.15$. Find the 5% critical region.

2

3

4. A single observation from $B(20, p)$ is used to test $H_0 : p = 0.4$ against $H_1 : p \neq 0.4$.

(a) Find the 5% critical region. (3 marks)

(b) Give the actual significance level. (1 mark)

a

b

5. A test statistic has distribution $B(20, p)$. Given $H_0 : p = 0.18$ and $H_1 : p < 0.18$, find the critical region using a 1% level of significance.

5

6. A random variable has distribution $B(10, p)$. A single observation is used to test $H_0 : p = 0.22$ against $H_1 : p \neq 0.22$.

(a) Using a 1% level of significance, find the critical region. The probability in each tail should be as close as possible to 0.005. (3 marks)

(b) Write down the actual significance level. (2 marks)

a

b

7. A component fails 3 times out of 10. An engineer tests 20 components made using a new system he thinks reduces failures.
(a) Describe the statistic. (1 mark)
(b) State hypotheses. (2 marks)
(c) Find the 5% critical region with probability as close to 0.05 as possible. (3 marks)
(d) Give the actual significance level. (1 mark)

a

b

c

d

8. An average of 12 seedlings in each tray of 36 survive. A gardener thinks a new fertiliser improves survival.
(a) Describe the statistic and give hypotheses. (3 marks)
(b) Find the 10% critical region. (3 marks)
(c) State the probability of incorrectly rejecting the null hypothesis. (1 mark)

a

b

c

Exercise 7B - Statistics 1 Pages 103-105

9. One fifth of customers choose lasagne. A restaurant owner changes the recipe and thinks the proportion will change.

(a) Suggest a model and hypotheses. (3 marks)

She samples 25 customers.

(b) Find the 5% critical region. (4 marks)

(c) Give the probability of incorrectly rejecting the null hypothesis. (1 mark)

a

b

c

Doing a full one-tailed hypothesis test

Miss Weir tosses a coin 8 times and it comes up heads 6 times. She claims the coin is **biased towards heads**. With a significance level of 5%, test her claim.

STEP 1: Define test statistic X (stating its distribution), and the parameter p.

STEP 2: Write null and alternative hypotheses.

STEP 3: Determine probability of observed test statistic (or 'more extreme'), assuming null hypothesis.
i.e. Determine probability we'd see this outcome just by chance.

STEP 4: Two-part conclusion:
1. Do we reject H_0 or not?
2. Put in context of original problem.

C.D.F. Binomial table: $p = 0.5, n = 8$

x	$P(X \leq x)$
0	0.0039
1	0.0352
2	0.1445
3	0.3633
4	0.6367
5	0.8555
6	0.9648
7	0.9961

NEW TO A LEVEL 2017: The probability of 'the observed value or more extreme' is known as the p-value.

Alternative method using critical regions 29

We can also find the critical region and see if the test statistic lies within it.

Miss Weir tosses a coin 8 times and it comes up heads 6 times. She claims the coin is **biased towards heads**. With a significance level of 5%, test her claim.

X is number of heads.
p is probability of heads.

$$X \sim B(8, p)$$

STEP 1: Define test statistic X (stating its distribution), and the parameter p.

STEP 2: Write null and alternative hypotheses.

$H_0 : p = 0.5$
$H_1 : p > 0.5$

STEP 3 (Alternative): Determine critical region.

STEP 4: Two-part conclusion:
1. Do we reject H_0 or not?
2. Put in context of original problem.

C.D.F. Binomial table: $p = 0.5, n = 8$

x	$P(X \leq x)$
0	0.0039
1	0.0352
2	0.1445
3	0.3633
4	0.6367
5	0.8555
6	0.9648
7	0.9961

More on p-values 30

Sheila wants to know if a coin is biased towards heads and throws it a large number of times, counting the number of heads. The p-value is less than 0.03.
Conduct a hypothesis test at the 5% significance level.

Ordinarily we'd calculate the probability of seeing the observed number of heads 'or more extreme'. But this has already been done for us (i.e. the p-value), so we just need to compare this against the threshold.

Further Example 31

[Textbook] The standard treatment for a particular disease has a $\frac{2}{5}$ probability of success. A certain doctor has undertaken research in this area and has produced a new drug which has been successful with 11 out of 20 patients. The doctor claims the new drug represents an improvement on the standard treatment. Test, at the 5% significance level, the claim made by the doctor.



STEP 1: Define test statistic X (stating its distribution), and the parameter p.

STEP 2: Write null and alternative hypotheses.

STEP 3: Determine probability of observed test statistic (or 'more extreme'), assuming null hypothesis.

STEP 4: Two-part conclusion:
1. Do we reject H_0 or not?
2. Put in context of original problem.

Test Your Understanding 32

Edexcel S2 Jan 2011 Q2

A student takes a multiple choice test. The test is made up of 10 questions each with 5 possible answers. The student gets 4 questions correct. Her teacher claims she was guessing the answers. Using a one tailed test, at the 5% level of significance, test whether or not there is evidence to reject the teacher's claim.
State your hypotheses clearly. **(6)**

Note the mark for stating distribution of X under null hypothesis.

Note two-mark conclusion.

Example 33

1. Jessica complains that her die is biased so that she is less likely to get a six. She tests this at the 5% significance level.

(i) Write down her null and alternative hypotheses.

Jessica throws the die 20 times and gets just one six.

(ii) Find the p-value for this test.

(iii) What is Jessica's conclusion?

(i)

(ii)

(iii)

Test your understanding 34

2. Hassan suspects his die is biased towards one. He tests at 10% and obtains 5 ones in 12 throws.

(i) Write down suitable hypotheses.

(ii) Find the p-value.

(iii) Give your conclusion.

(i)

(ii)

(iii)

Practice.. 35

3. A bus company claims its service is on time in 90% of journeys. I suspect it is lower and test at 5%. In 15 journeys the bus is on time 10 times.
(i) Write down suitable hypotheses.
(ii) Find the p-value.
(iii) Give your conclusion.

(i)

(ii)

(iii)

.. and more practice. 36

4. A bus service was on time in 80% of journeys. The company claims improvements increased punctuality. I test at 10%. In 16 journeys the bus is on time 15 times.
(i) Write down suitable hypotheses.
(ii) Find the p-value.
(iii) Give your conclusion.

(i)

(ii)

(iii)

1. A single observation $x = 5$ is obtained from $B(10, p)$. At the 5% level test $H_0 : p = 0.25$ against $H_1 : p > 0.25$.
2. A single observation $x = 1$ is obtained from $B(10, p)$. At the 5% level test $H_0 : p = 0.4$ against $H_1 : p < 0.4$.

1

2

3. A single observation $x = 10$ is obtained from $B(20, p)$. At the 5% level test $H_0 : p = 0.3$ against $H_1 : p > 0.3$.
4. A single observation $x = 3$ is obtained from $B(20, p)$. At the 1% level test $H_0 : p = 0.45$ against $H_1 : p < 0.45$.

3

4

5. A single observation $x = 2$ is obtained from $B(20, p)$. At the 5% level test $H_0 : p = 0.28$ against $H_1 : p < 0.28$.

6. A single observation $x = 7$ is obtained from $B(8, p)$. At the 5% level test $H_0 : p = 0.32$ against $H_1 : p > 0.32$.

5

6

7. A board-game die is suspected of giving too few sixes. One six appears in 12 rolls. Test at 5% whether the probability of a six is less than one sixth.

7

8. A skin treatment has a claimed success rate of 68%.
(a) Out of n patients, X successes occur. Give and justify a model.
Only 3 of 10 randomly sampled patients recover. It is thought the treatment is less effective than claimed.
(b) Test at 5%.

a

b

Two-Tailed Tests 42

We have already seen that if we're interest in bias 'either way', we have two tails, and therefore have to split the critical region by **halving the significance level at each end**.

Over a long period of time it has been found that in Enrico's restaurant the ratio of non-veg to veg meals is 2 to 1. In Manuel's restaurant in a random sample of 10 people ordering meals, 1 ordered a vegetarian meal. Using a 5% level of significance, test whether or not the proportion of people eating veg meals in Manuel's restaurant is different to that in Enrico's restaurant.

Half significance as 2 tailed.

Conclusion and what it means in context.

Test Your Understanding 43

Edexcel S2 Jan 2006 Q7a

A teacher thinks 20% of pupils in a school read the Deano comic regularly. He chooses 20 pupils at random and finds that 9 read the Deano.

(a)(i) Test at the 5% significance level whether the percentage differs from 20%. State your hypotheses clearly.

(ii) State all possible numbers of Deano readers from a sample of 20 that make the test significant at the 5% level. (9 marks)

(a)(i)

(ii)

Exercise 7D - Statistics 1 Pages 108-109 44

1. A single observation $x = 10$ is obtained from $B(30, p)$. At the 5% level test $H_0 : p = 0.5$ against $H_1 : p \neq 0.5$.
2. A single observation $x = 10$ is obtained from $B(25, p)$. At the 10% level test $H_0 : p = 0.3$ against $H_1 : p \neq 0.3$.

1

2

3. A single observation $x = 9$ is obtained from $B(10, p)$. At the 5% level test $H_0 : p = 0.75$ against $H_1 : p \neq 0.75$.
4. A single observation $x = 1$ is obtained from $B(20, p)$. At the 1% level test $H_0 : p = 0.6$ against $H_1 : p \neq 0.6$.

3

4

5. A single observation $x = 4$ is obtained from $B(50, p)$. At the 2% level test $H_0 : p = 0.02$ against $H_1 : p \neq 0.02$.
6. A coin lands heads 6 times in 20 tosses. Use a two-tailed 5% test to decide whether there is evidence of bias.

5

6

7. Nationally 20% of patients experience complications after an operation. A hospital samples 20 records.
(a) Find critical regions for a two-tailed 5% test of whether its rate differs. Take each tail as close to 2.5% as possible. (5 marks)
(b) State the actual significance level. (1 mark)
(c) Eight patients had complications. Comment using the critical regions. (2 marks)

a

b

c

8. One in ten glass bowls has a hairline crack. After modifying the process, one of 20 sampled bowls is cracked. Test at 10% whether the proportion has changed. State your hypotheses. (7 marks)

8

p-value

9. Agnetha's carrots have a 25% chance of being longer than 7 cm. After she changes fertiliser, 13 of 30 randomly sampled carrots exceed 7 cm. Test her claim that the probability has changed at 5%. State your hypotheses. (7 marks)

9

10. A standard blood test correctly diagnoses a disease with probability 0.96. A cheaper test is claimed to have the same probability. In a clinical trial it diagnoses 63 of 75 patients correctly. Test the claim at 10%, stating your hypotheses. (7 marks)

10